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Association of Dipeptidyl Peptidase-4 Inhibitors with Glaucoma Risk in Patients with Type 2 Diabetes: A Nationwide Cohort Study

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Purpose: To investigate the association between dipeptidyl peptidase-4 inhibitors (DPP4is) and the risk of primary open-angle glaucoma (POAG) and normal tension glaucoma (NTG) in patients with type 2 diabetes mellitus (T2DM).

Design: Retrospective cohort study.

Subjects: A total of 582 710 patients with T2DM treated with either DPP4i (exposure group) or non-DPP4i medications (control group) were analyzed between 2008 and 2021.

Methods: Patients were 1-to-1 matched by propensity scores on demographic and clinical characteristics. Cox proportional hazards models were applied to estimate hazard ratios for POAG and NTG, adjusting for age, sex, comorbidities, and concurrent antidiabetic medications.

Main Outcome Measures: Incidences of POAG and NTG.

Results: Dipeptidyl peptidase-4 inhibitor users demonstrated a significantly lower risk of POAG (adjusted hazard ratio [aHR], 0.53; 95% confidence interval [CI], 0.50–0.56) and NTG (aHR, 0.55; 95% CI, 0.50–0.62) compared to non-DPP4i users on first-generation diabetes medication. Subgroup analysis revealed a consistent risk reduction across all age groups (18–39: aHR, 0.56; 95% CI, 0.51–0.62; 40–64: aHR, 0.52; 95% CI, 0.47–0.57; ≥65 years: aHR, 0.51; 95% CI, 0.47–0.56) and among patients with or without diabetic-related complications, including diabetic retinopathy, diabetic kidney disease, and diabetic neuropathy (DN) (aHR: without vs. with diabetic retinopathy [0.54 vs. 0.43], without vs. with diabetic kidney disease [0.53 vs. 0.49], without vs. with DN [0.54 vs.0.43]), with all comparisons showing statistical significance ($P < 0.001$). Cumulative incidence analyses revealed a sustained lower risk for DPP4i users throughout the study period (log-rank $P < 0.001$).

Conclusions: Exposure to DPP4i was associated with a reduced risk of developing POAG and NTG compared with users of first-generation diabetes medication. Further research is needed to explore the underlying mechanisms and their implications for glaucoma prevention and management.

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Diabetes poses a significant public health concern, affecting 10.5% of adults worldwide, with type 2 diabetes mellitus (T2DM) accounting for >90% of cases. The high prevalence of T2DM significantly contributes to morbidity, including an increased risk of glaucoma. Glaucoma, particularly primary open-angle glaucoma (POAG) and normal tension glaucoma (NTG), further exacerbates the chronic disease burden associated with T2DM.[1–3]

Antidiabetic medications provide cardiovascular, renal, and neurotrophic benefits beyond glucose control.[4] Metformin—the preferred first-line therapy for T2DM[5]—has shown geroprotective properties and a potential to reduce the risk of open-angle glaucoma compared with other first-generation diabetes drugs.[6] Recent retrospective studies have extended these findings to second-generation diabetes drugs like glucagon-like peptide-1 receptor agonists (GLP-1 RAs)[7,8] and sodium-glucose cotransporter 2 inhibitors,[9] both of which are associated with lower glaucoma risk. Furthermore, GLP-1 RAs, which act on incretin system, have shown various systemic advantages, including potential neuroprotective properties.[10,11]

Dipeptidyl peptidase-4 inhibitors (DPP4is), another second-generation diabetes drug, prolong endogenous incretin activity by inhibiting the rapid degradation of incretins.[10] While DPP4is share incretin-mediated effects with GLP-1 RAs, emerging evidence suggests that DPP4is may confer additional benefits through mechanisms independent of incretins.[12–14] These may involve modulation of reactive oxygen species (ROS) and glutamate signaling, potentially contributing to optic nerve health in ways that differ from GLP-1 RAs.[12–15] Rodent studies have demonstrated DPP4i benefits in inflammatory, microvascular, and

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neurodegenerative disorders, such as Parkinson disease[11] and cerebral ischemia.[16] However, limited studies have explored the association between DPP4is and glaucoma. A recent study found lower glaucoma risk with sodium-glucose cotransporter 2 inhibitors compared with DPP4is, but it lacked comparisons with other antidiabetic medications and did not adjust for concurrent use of other glucose-lowering agents, which may affect glaucoma risk.[17]

Given the emerging evidence linking antidiabetic agents to glaucoma risk reduction, DPP4is represent a promising candidate due to its incretin-mediated and independent effects. This study aims to investigate the association between DPP4i use and the risk of developing POAG or NTG in patients with T2DM. To minimize confounding, we excluded users of other second-generation diabetes drugs previously associated with glaucoma risk. Our analysis specifically focused on evaluating the association between DPP4i and first-generation diabetes medication (including metformin, sulfonylureas, thiazolidinediones, α-glucosidase inhibitors, and glinide).

Methods

Data Source

This retrospective cohort study analyzed data from Taiwan's National Health Insurance Research Database, a database covering >99% of the population. Diagnoses in the National Health Insurance Research Database are coded using the International Classification of Diseases 9 and 10, Clinical Modification (ICD9-CM and ICD10-CM). All analyses were conducted within a secure environment at the Health and Welfare Data Science Center under the supervision of the Ministry of Health and Welfare. Ethical approval for the study was granted by the Research Ethics Committee of China Medical University Hospital (CMUH110-REC1-038[CR-4]).

Study Population

Eligible patients were adults diagnosed with T2DM, identified using ICD-9-CM codes 250.x0 and 250.x2 or ICD-10-CM code E11. To improve case ascertainment, patients were required to have ≥3 outpatient claims or ≥1 inpatient claim for T2DM. Patients were included if they were prescribed either DPP4is (Anatomical Therapeutic Chemical codes: A10BH01–A10BH05, A10BH09) or other diabetes medications (listed in Table S1, available at www.ophthalmologyscience.org) for a minimum of 3 months. The DPP4i cohort comprised patients initiating DPP4i therapy, with the index date defined as the date of their first DPP4i prescription.

To establish a comparable cohort and ensure a consistent observation period, non-DPP4i users were matched 1:1 with DPP4i users using propensity score matching based on variables such as age, sex, index year, dwelling region, urbanization level, insurance premium, comorbidities, and Charlson Comorbidity Index scores. The index date for each non-DPP4i user was assigned to match the index year of their corresponding DPP4i user.

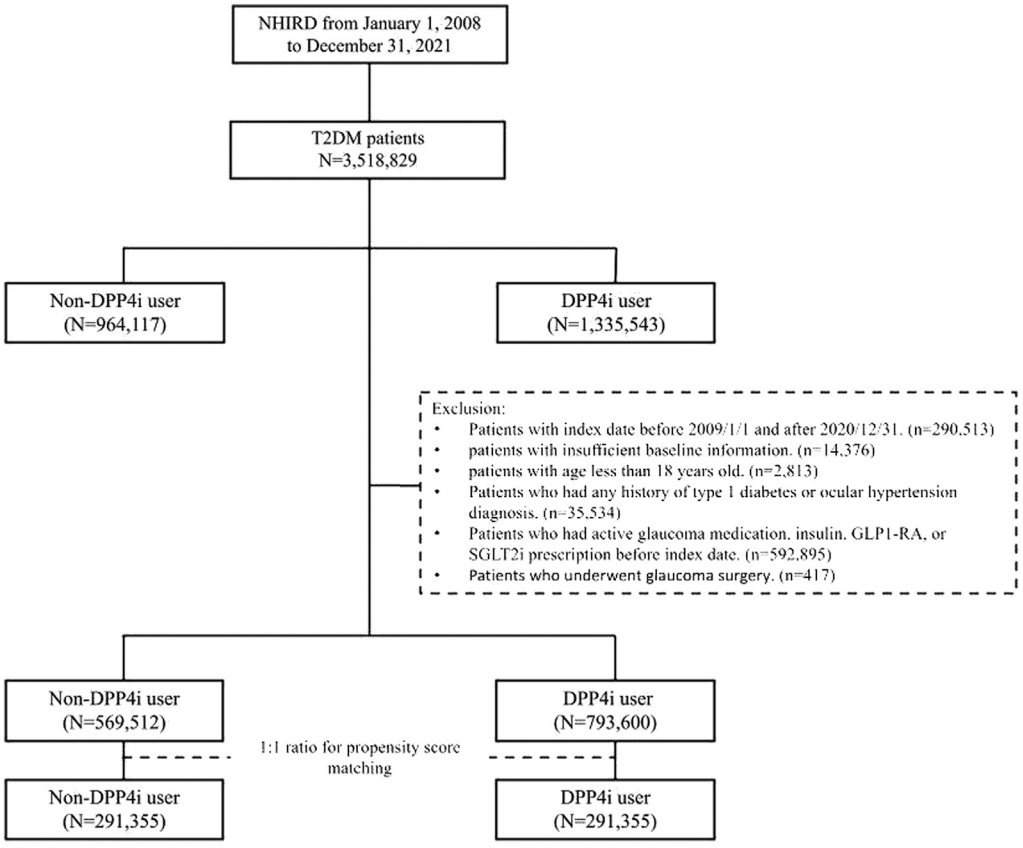


Figure 1. Flowchart of study participants. From the NHIRD, 582 710 patients with type 2 diabetes were included, comprising 291 355 DPP4i users and 291 355 non-DPP4i users matched by propensity scores. DPP4i = dipeptidyl peptidase-4 inhibitor; GLP-1 RA = glucagon-like peptide-1 receptor agonist; NHIRD = National Health Insurance Research Database; T2DM = type 2 diabetes mellitus.

Table 2. Baseline Characteristics in T2DM between DPP4i and Non-DPP4i User

Variable	Non-DPP4i User (*N* = 291 355)		DPP4i User (*N* = 291 355)		SMD
	n	%	*n*	%	
Sex					0.003
Female	128 994	44.27	129 425	44.42	
Male	162 361	55.73	161 930	55.58	
Age (yrs)					
18–39	90 025	30.90	89 907	30.86	0.001
40–64	85 271	29.27	85 912	29.49	0.005
≥65	116 059	39.83	115 536	39.65	0.004
Mean ± SD	60.58	12.67	60.49	12.59	0.007
Region of dwelling					
Northern	129 826	44.56	128 010	43.94	0.013
Central	68 593	23.54	68 171	23.40	0.003
Southern	84 313	28.94	86 794	29.79	0.019
Eastern	8623	2.96	8380	2.88	0.005
Degree of urbanization					
Lowest	27 597	9.47	27 228	9.35	0.004
Median	112 405	38.58	112 807	38.72	0.003
Highest	151 353	51.95	151 320	51.94	<0.001
Insurance premium (NTD per month)					
<18 000	74 172	25.46	72 597	24.92	0.012
18 000–34 999	141 257	48.48	142 764	49.00	0.010
≥35 000	75 926	26.06	75 994	26.08	0.001
Comorbidities					
Coronary artery disease	62 012	21.28	60 150	20.64	0.016
Stroke	36 564	12.55	35 013	12.02	0.016
Peripheral artery disease	4378	1.50	4227	1.45	0.004
Atrial fibrillation	26 688	9.16	25 624	8.79	0.013
Heart failure	12 738	4.37	12 212	4.19	0.009
Hypertension	195 969	67.26	196 614	67.48	0.005
Dyslipidemia	185 583	63.70	186 489	64.01	0.006
CKD	17 340	5.95	17 514	6.01	0.003
COPD	31 052	10.66	30 107	10.33	0.011
Depression	18 751	6.44	18 262	6.27	0.007
Schizophrenia	3156	1.08	3224	1.11	0.002
Rheumatoid diseases	5387	1.85	5308	1.82	0.002
Osteoporosis	14 973	5.14	14 674	5.04	0.005
Alcoholism	5780	1.98	5571	1.91	0.005
Nicotine dependence	8357	2.87	8134	2.79	0.005
Cancer	14 732	5.06	13 830	4.75	0.014
Retinopathy	23 733	8.15	25 518	8.76	0.022
Myopia	2690	0.92	2606	0.89	0.003
Cataract	63 530	21.81	63 296	21.72	0.002
Uveitis	1814	0.62	1686	0.58	0.006
DR	10 352	3.55	12 706	4.36	0.041
DKD	21 045	7.22	22 943	7.87	0.025
DN	13 802	4.74	15 494	5.32	0.027
CCI score					
0	225 423	77.37	228 458	78.41	0.025
1	31 235	10.72	30 077	10.32	0.013
≥2	34 697	11.91	32 820	11.26	0.020
Number of DM medication					
Mean ± SD	2.11	0.76	2.1	0.77	0.009

CCI = Charlson Comorbidity Index; CKD = chronic kidney disease; COPD = chronic obstructive pulmonary disease; DKD = diabetic kidney disease; DM = diabetes mellitus; DN = diabetic neuropathy; DPP4i = dipeptidyl peptidase-4 inhibitor; DR = diabetic retinopathy; NTD = New Taiwan Dollar; SD = standard deviation; SMD = standardized mean differences; T2DM = type 2 diabetes mellitus.

Additionally, matching considered the number of diabetes medication classes prescribed within 45 days of the index date. For example, if a patient initiating DPP4i therapy was already on 1 class of diabetes medication, the matched patient in the non-DPP4i cohort was also required to initiate a second non-DPP4i diabetes medication class.

Table 3. Hazard Ratios of POAG among T2DM with DPP4i and Non-DPP4i User

Variable	POAG			Crude		Adjusted*	
	N	*PY*	*IR*	*HR (95% CI)*	P *Value*	*HR (95% CI)*	P *Value*
Non-DPP4i user	6128	1 628 190.67	3.76	1.00 (reference)	-	1.00 (reference)	-
DPP4i user	3159	1 759 130.36	1.8	0.48 (0.46, 0.50)§	<0.001	0.53 (0.50, 0.56)§	<0.001
Sex							
Female	4039	1 542 610.63	2.62	1.00 (reference)	-	1.00 (reference)	-
Male	5248	1 844 710.41	2.84	1.09 (1.04, 1.13)§	<0.001	1.14 (1.09, 1.19)§	<0.001
Age							
18–39	2680	1 097 550.35	2.44	1.00 (reference)	-	1.00 (reference)	-
40–64	2842	1 029 490.34	2.76	1.13 (1.07, 1.19)§	<0.001	1.08 (1.03, 1.14)‡	0.004
≥65	3765	1 260 280.34	2.99	1.23 (1.17, 1.29)§	<0.001	1.12 (1.05, 1.18)§	<0.001
Region of dwelling							
Northern	4604	1 532 851.05	3	1.00 (reference)	-	1.00 (reference)	-
Central	2080	784 404.68	2.65	0.88 (0.84, 0.93)§	<0.001	0.90 (0.85, 0.95)§	<0.001
Southern	2370	971 888.85	2.44	0.81 (0.77, 0.85)§	<0.001	0.82 (0.78, 0.87)§	<0.001
Eastern	233	98 176.45	2.37	0.79 (0.69, 0.90)§	<0.001	0.81 (0.71, 0.92)‡	0.001
Degree of urbanization							
Lowest	719	315 518.46	2.28	1.00 (reference)	-	1.00 (reference)	-
Median	3462	1 302 505.88	2.66	1.17 (1.08, 1.26)§	<0.001	1.17 (1.08, 1.27)§	<0.001
Highest	5106	1 769 296.70	2.89	1.27 (1.17, 1.37)§	<0.001	1.23 (1.14, 1.33)§	<0.001
Insurance premium (NTD per month)							
<18 000	2501	850 601.33	2.94	1.00 (reference)	-	1.00 (reference)	-
18 000–34999	4262	1 645 882.03	2.59	0.88 (0.84, 0.92)§	<0.001	0.93 (0.88, 0.98)‡	0.005
≥35 000	2524	890 837.68	2.83	0.96 (0.91, 1.02)	0.181	0.98 (0.92, 1.03)	0.430
Comorbidities							
Coronary artery disease							
No	7306	2 707 965.44	2.7	1.00 (reference)	-	1.00 (reference)	-
Yes	1981	679 355.59	2.92	1.08 (1.03, 1.14)‡	0.002	1.09 (1.04, 1.15)‡	0.001
Stroke							
No	8275	3 014 968.45	2.74	1.00 (reference)	-	1.00 (reference)	-
Yes	1012	372 352.58	2.72	0.99 (0.93, 1.06)	0.801	0.98 (0.91, 1.05)	0.509
Peripheral artery disease							
No	9177	3 341 777.20	2.75	1.00 (reference)	-	1.00 (reference)	-
Yes	110	45 543.84	2.42	0.88 (0.73, 1.06)	0.184	0.83 (0.68, 1.00)†	0.048
Atrial fibrillation							
No	8548	3 118 458.50	2.74	1.00 (reference)	-	1.00 (reference)	-
Yes	739	268 862.53	2.75	1.00 (0.93, 1.08)	0.922	1.00 (0.93, 1.08)	0.955
Heart failure							
No	9002	3 264 911.80	2.76	1.00 (reference)	-	1.00 (reference)	-
Yes	285	122 409.23	2.33	0.85 (0.75, 0.95)‡	0.006	0.86 (0.76, 0.97)†	0.016
Hypertension							
No	3126	1 142 074.35	2.74	1.00 (reference)	-	1.00 (reference)	-
Yes	6161	2 245 246.69	2.74	1.00 (0.96, 1.05)	0.927	0.98 (0.94, 1.03)	0.488
Dyslipidemia							
No	3557	1 269 263.53	2.8	1.00 (reference)	-	1.00 (reference)	-
Yes	5730	2 118 057.50	2.71	0.96 (0.92, 1.01)	0.087	0.94 (0.90, 0.98)**	0.003
CKD							
No	8899	3 229 270.76	2.76	1.00 (reference)	-	1.00 (reference)	-
Yes	388	158 050.28	2.45	0.89 (0.81, 0.99)†	0.030	0.88 (0.79, 0.98)†	0.019
COPD							
No	8456	3 078 773.39	2.75	1.00 (reference)	-	1.00 (reference)	-
Yes	831	308 547.64	2.69	0.98 (0.91, 1.05)	0.617	0.96 (0.90, 1.04)	0.331
Depression							
No	8725	3 196 017.31	2.73	1.00 (reference)	-	1.00 (reference)	-
Yes	562	191 303.72	2.94	1.08 (0.99, 1.17)	0.090	1.11 (1.01, 1.21)†	0.025
Schizophrenia							
No	9192	3 352 577.92	2.74	1.00 (reference)	-	1.00 (reference)	-
Yes	95	34 743.11	2.73	1.00 (0.82, 1.22)	1.000	1.05 (0.85, 1.29)	0.651
Rheumatoid diseases							
No	9136	3 333 248.84	2.74	1.00 (reference)	-	1.00 (reference)	-
Yes	151	54 072.19	2.79	1.02 (0.87, 1.20)	0.811	1.04 (0.89, 1.23)	0.607
Osteoporosis							
No	8878	3 229 792.55	2.75	1.00 (reference)	-	1.00 (reference)	-
Yes	409	157 528.49	2.6	0.95 (0.86, 1.04)	0.276	0.92 (0.83, 1.02)	0.097

Table 3. (Continued.)

Variable	POAG			Crude		Adjusted*	
	N	*PY*	*IR*	*HR (95% CI)*	P *Value*	*HR (95% CI)*	P *Value*
Alcoholism							
No	9173	3 332 001.00	2.75	1.00 (reference)	-	1.00 (reference)	-
Yes	114	55 320.04	2.06	0.75 (0.62, 0.90)‡	0.002	0.79 (0.66, 0.95)†	0.014
Nicotine dependence							
No	9139	3 316 582.26	2.76	1.00 (reference)	-	1.00 (reference)	-
Yes	148	70 738.77	2.09	0.76 (0.65, 0.90)‡	0.001	0.78 (0.67, 0.92)‡	0.003
Cancer							
No	8906	3 246 943.29	2.74	1.00 (reference)	-	1.00 (reference)	-
Yes	381	140 377.75	2.71	0.99 (0.90, 1.10)	0.910	1.16 (1.00, 1.33)†	0.044
Retinopathy							
No	8083	3 106 238.86	2.6	1.00 (reference)	-	1.00 (reference)	-
Yes	1204	281 082.17	4.28	1.65 (1.55, 1.75)§	<0.001	1.35 (1.24, 1.47)§	<0.001
Myopia							
No	9162	3 360 536.52	2.73	1.00 (reference)	-	1.00 (reference)	-
Yes	125	26 784.51	4.67	1.71 (1.44, 2.04)§	<0.001	1.61 (1.35, 1.92)§	<0.001
Cataract							
No	6713	2 684 783.88	2.5	1.00 (reference)	-	1.00 (reference)	-
Yes	2574	702 537.16	3.66	1.47 (1.40, 1.53)§	<0.001	1.39 (1.32, 1.47)§	<0.001
Uveitis							
No	9219	3 369 797.95	2.74	1.00 (reference)	-	1.00 (reference)	-
Yes	68	17 523.08	3.88	1.42 (1.12, 1.80)‡	0.004	1.19 (0.94, 1.51)	0.157
DR							
No	8662	3 247 469.72	2.67	1.00 (reference)	-	1.00 (reference)	-
Yes	625	139 851.31	4.47	1.67 (1.54, 1.82)§	<0.001	1.09 (0.97, 1.21)	0.155
DKD							
No	8615	3 156 898.16	2.73	1.00 (reference)	-	1.00 (reference)	-
Yes	672	230 422.88	2.92	1.07 (0.99, 1.16)	0.094	1.02 (0.94, 1.11)	0.6
DN							
No	8652	3 210 264.07	2.7	1.00 (reference)	-	1.00 (reference)	-
Yes	635	177 056.96	3.59	1.33 (1.23, 1.44)§	<0.001	1.24 (1.15, 1.35)§	<0.001
CCI score							
0	7637	2 731 688.32	2.8	1.00 (reference)	-	1.00 (reference)	-
1	839	327 369.00	2.56	0.92 (0.85, 0.98)†	0.016	0.90 (0.83, 0.96)‡	0.003
≥2	811	328 263.71	2.47	0.89 (0.82, 0.95)‡	0.001	0.82 (0.74, 0.91)§	<0.001

CCI = Charlson Comorbidity Index; CI = confidence interval; CKD = chronic kidney disease; COPD = chronic obstructive pulmonary disease; DKD = diabetic kidney disease; DN = diabetic neuropathy; DPP4i = dipeptidyl peptidase-4 inhibitor; DR = diabetic retinopathy; HR = hazard ratio; IR = incidence rate per 1000 person-years; NTD = New Taiwan Dollar; POAG = primary open angle glaucoma; PY = person-years; T2DM = type 2 diabetes mellitus.
*Adjusted HR: adjusted by sex, age, region of dwelling, degree of urbanization, insurance premium, comorbidities, CCI score, and medication.
†*P* value <0.05.
‡*P* < 0.01.
§*P* < 0.001.

Patients with insulin use prior to the index date were excluded to minimize potential confounding, as they often represent a population with more advanced or poorly controlled diabetes, which could independently affect glaucoma risk. Additional exclusion criteria included incomplete baseline data, age <18 years, a history of type 1 diabetes or ocular hypertension, prior glaucoma surgery, and any use of glaucoma medications, GLP-1 RAs, or sodium-glucose cotransporter 2 inhibitors before the index date. The study enrollment period spanned from January 1, 2009, to December 31, 2020, with follow-up until December 31, 2021. A flowchart illustrating patient selection is presented in Figure 1.

Outcomes and Covariates

The primary outcomes were the incidence of POAG and NTG. Primary open-angle glaucoma was identified using ICD-9-CM codes 365.10 to 365.12, 365.15, and ICD-10-CM codes H40.10 to H40.12, H40.15. Normal tension glaucoma was defined using ICD-9-CM code 365.12 and ICD-10-CM code H40.12. To enhance diagnostic accuracy, a diagnosis of POAG or NTG required ≥2 outpatient claims or 1 inpatient claim with the corresponding codes. Follow-up continued until the first occurrence of POAG, NTG, death, or December 31, 2021.

Baseline characteristics included demographic factors (age, sex, and region of dwelling), socioeconomic status (insurance premium levels and urbanization level), and clinical factors (comorbidities such as hypertension, dyslipidemia, coronary artery disease, stroke, chronic kidney disease, retinopathy, cataract, uveitis, myopia, diabetic retinopathy, diabetic kidney disease, and diabetic neuropathy [DN]), as shown in Table 2. The use of first-generation diabetes medication—including metformin, sulfonylureas, thiazolidinediones, α-glucosidase inhibitors, and glinides—were considered as covariates in the analyses.

Statistical Analysis

Baseline characteristics were summarized using means and standard deviations for continuous variables and counts with percentages for categorical variables. Standardized mean differences were calculated to assess the balance between the DPP4i and non-DPP4i cohorts after propensity score matching, with standardized mean differences <0.1 indicating minimal imbalance. Cox proportional hazards models were employed to estimate hazard ratios and 95% confidence intervals (CIs) for the development of POAG and NTG. Both unadjusted and adjusted models were constructed, with adjustments made for potential confounders such as age, sex, comorbidities, and concurrent medication use.

Cumulative incidences of POAG and NTG were estimated using the Kaplan–Meier method, and differences between the DPP4i and non-DPP4i cohorts were assessed using the log-rank test. Statistical significance was set at a 2-sided P value of <0.05. All analyses were performed using SAS version 9.4 (SAS Inc) and R software version 4.4.1 (R Foundation for Statistical Computing).

Results

Baseline Characteristics

After propensity score matching, a total of 291 355 individuals were included in both the DPP4i users and non-DPP4i users cohorts, resulting in a combined study population of 582 710 patients with T2DM. Baseline characteristics were well-balanced across demographic, socioeconomic, and clinical factors, with standardized mean differences below 0.1 for all variables, demonstrating comparability between groups (Table 2). The mean age in both cohorts was approximately 60 years, with a male predominance (about 56% in both). Comorbidities such as hypertension (67.26% in non-DPP4i group vs. 67.48% in DPP4i group), dyslipidemia (63.70% in non-DPP4i vs. 64.01% in DPP4i), cataract (21.81% in non-DPP4i vs. 21.72% in DPP4i), diabetic retinopathy (3.55% in non-DPP4i vs. 4.36% in DPP4i), diabetic kidney disease (7.22% in non-DPP4i vs. 7.87% in DPP4i), and DN (4.74% in non-DPP4i vs. 5.32% in DPP4i), were similarly distributed between the cohorts.

DPP4i Use and Risk of POAG

During a mean follow-up period of 6.04 ± 3.25 years for DPP4i users and 5.59 ± 3.39 years for non-DPP4i users, a total of 3159 cases of POAG were identified in the DPP4i users cohort and 6128 cases in the non-DPP4i users cohort. Dipeptidyl peptidase-4 inhibitor use was associated with a significantly lower risk of developing POAG compared with non-DPP4i use (adjusted hazard ratio [aHR], 0.53; 95% CI, 0.50–0.56; $P < 0.001$) (Table 3). Male sex was associated with a slightly higher risk of POAG compared with female sex (aHR, 1.14; 95% CI, 1.09–1.19; $P < 0.001$). Advanced age and certain comorbidities, including coronary artery disease, depression, cancer, retinopathy, myopia, cataract, and DN, were linked to an increased risk of POAG (Table 3). The cumulative incidence of POAG was consistently lower in DPP4i users compared with non-DPP4i users over the study period, as illustrated by the Kaplan–Meier survival curves (Fig 2). The log-rank test indicated a significant difference between the cohorts ($P < 0.001$).

POAG Risk across Different Subgroups

Subgroup analysis was conducted to assess the consistency of the association between DPP4i use and POAG risk across various demographic and clinical subgroups (Table 4). Dipeptidyl peptidase-4 inhibitor use was linked to a reduced risk of POAG across various demographic factors, including both men (aHR, 0.54; 95% CI, 0.51–0.58; $P < 0.001$) and women (aHR, 0.51; 95% CI, 0.47–0.55; $P < 0.001$). Notably, the risk reduction became more pronounced with increasing age (aged 18–39: aHR, 0.56; 95% CI, 0.51–0.62; aged 40–64: aHR, 0.52; 95% CI, 0.47–0.57; aged ≥65 years: aHR, 0.51; 95% CI, 0.47–0.56). Among patients with comorbid conditions, DPP4i users consistently showed a lower risk of POAG, with aHR of 0.53 (95% CI, 0.47–0.59; $P < 0.001$) for coronary artery disease, 0.57 (95% CI, 0.48–0.66; $P < 0.001$) for stroke, 0.45 (95% CI, 0.27–0.75; $P = 0.002$) for peripheral artery disease, 0.58 (95% CI, 0.48–0.69; $P < 0.001$) for atrial fibrillation, 0.43 (95% CI, 0.32–0.59; $P < 0.001$) for heart failure, 0.52 (95% CI, 0.49–0.56; $P < 0.001$) for hypertension, 0.54 (95% CI, 0.50–0.57; $P < 0.001$) for dyslipidemia, 0.44 (95% CI, 0.34–0.57; $P < 0.001$) for chronic kidney disease, 0.52 (95% CI, 0.43–0.62; $P < 0.001$) for chronic obstructive pulmonary disease, 0.53 (95% CI, 0.42–0.65; $P < 0.001$) for depression, 0.40 (95% CI, 0.26–0.61; $P < 0.001$) for rheumatoid diseases, 0.47 (95% CI, 0.36–0.60; $P < 0.001$) for osteoporosis, 0.54 (95% CI, 0.36–0.81; $P = 0.003$) for nicotine dependence, 0.51 (95% CI, 0.39–0.66; $P < 0.001$) for

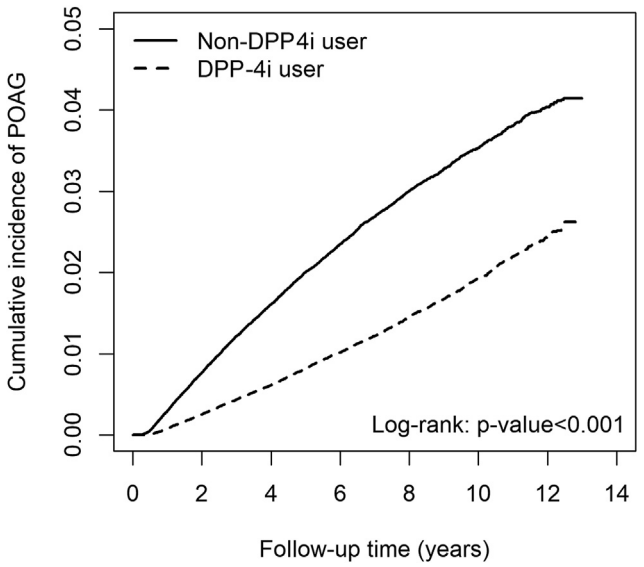


Figure 2. The cumulative incidence curves of POAG in DPP4i and non-DPP4i users. Kaplan–Meier curves showing a significantly lower cumulative incidence of POAG in DPP4i users compared with non-DPP4i users. DPP4i = dipeptidyl peptidase-4 inhibitor; POAG = primary open-angle glaucoma.

Table 4. Stratified Analysis of Different Covariates for the Risk of POAG

Variable	POAG in Non-DPP4i User			POAG in DPP4i User			Crude		Adjusted*	
	n	*PY*	*IR*	*n*	*PY*	*IR*	*HR (95% CI)*	P *Value*	*HR (95% CI)*	P *Value*
Sex										
Female	2688	743 042.61	3.62	1351	799 568.02	1.69	0.47 (0.44, 0.50)§	<0.001	0.51 (0.47, 0.55)§	<0.001
Male	3440	885 148.07	3.89	1808	959 562.34	1.88	0.48 (0.46, 0.51)§	<0.001	0.54 (0.51, 0.58)§	<0.001
Age										
18–39	1765	532 619.79	3.31	915	564 930.56	1.62	0.49 (0.45, 0.53)§	<0.001	0.56 (0.51, 0.62)§	<0.001
40–64	1861	493 652.67	3.77	981	535 837.68	1.83	0.48 (0.45, 0.52)§	<0.001	0.52 (0.47, 0.57)§	<0.001
≥65	2502	601 918.22	4.16	1263	658 362.12	1.92	0.46 (0.43, 0.49)§	<0.001	0.51 (0.47, 0.56)§	<0.001
Region of dwelling										
Northern	2988	744 609.20	4.01	1616	788 241.85	2.05	0.51 (0.48, 0.54)§	<0.001	0.57 (0.53, 0.62)§	<0.001
Central	1409	378 788.79	3.72	671	405 615.89	1.65	0.44 (0.40, 0.48)§	<0.001	0.48 (0.43, 0.53)§	<0.001
Southern	1584	457 844.23	3.46	786	514 044.62	1.53	0.44 (0.40, 0.48)§	<0.001	0.49 (0.44, 0.55)§	<0.001
Eastern	147	46 948.45	3.13	86	51 228.00	1.68	0.53 (0.41, 0.70)§	<0.001	0.52 (0.38, 0.71)§*	<0.001
Degree of urbanization										
Lowest	491	152 788.39	3.21	228	162 730.07	1.40	0.43 (0.37, 0.51)§	<0.001	0.50 (0.41, 0.60)§	<0.001
Median	2310	621 874.46	3.71	1152	680 631.42	1.69	0.45 (0.42, 0.49)§	<0.001	0.51 (0.47, 0.56)§	<0.001
Highest	3327	853 527.83	3.9	1779	915 768.87	1.94	0.50 (0.47, 0.53)§	<0.001	0.54 (0.51, 0.58)§	<0.001
Insurance premium (NTD per month)										
<18 000	1639	409 038.16	4.01	862	441 563.16	1.95	0.48 (0.45, 0.53)§	<0.001	0.53 (0.48, 0.58)§	<0.001
18 000–34999	2869	789 948.13	3.63	1393	855 933.90	1.63	0.45 (0.42, 0.48)§	<0.001	0.50 (0.47, 0.54)§	<0.001
≥35 000	1620	429 204.38	3.77	904	461 633.30	1.96	0.52 (0.48, 0.56)§	<0.001	0.58 (0.53, 0.64)§	<0.001
Comorbidities										
Coronary artery disease										
No	4814	1 304 795.78	3.69	2492	1 403 169.67	1.78	0.48 (0.46, 0.50)§	<0.001	0.53 (0.50, 0.56)§	<0.001
Yes	1314	323 394.90	4.06	667	355 960.69	1.87	0.46 (0.42, 0.50)§	<0.001	0.53 (0.47, 0.59)§	<0.001
Stroke										
No	5492	1 449 798.52	3.79	2783	1 565 169.94	1.78	0.47 (0.45, 0.49)§	<0.001	0.52 (0.50, 0.55)§	<0.001
Yes	636	178 392.16	3.57	376	193 960.42	1.94	0.54 (0.47, 0.61)§	<0.001	0.57 (0.48, 0.66)§	<0.001
Peripheral artery disease										
No	6054	1 605 724.27	3.77	3123	1 736 052.93	1.80	0.48 (0.46, 0.50)§	<0.001	0.53 (0.50, 0.56)§	<0.001
Yes	74	22 466.40	3.29	36	23 077.43	1.56	0.47 (0.32, 0.70)§	<0.001	0.45 (0.27, 0.75)‡	0.002
Atrial fibrillation										
No	5651	1 498 667.50	3.77	2897	1 619 791.00	1.79	0.47 (0.45, 0.49)§	<0.001	0.52 (0.50, 0.55)§	<0.001
Yes	477	129 523.18	3.68	262	139 339.36	1.88	0.51 (0.44, 0.59)§	<0.001	0.58 (0.48, 0.69)§	<0.001
Heart failure										
No	5942	1 569 937.63	3.78	3060	1 694 974.17	1.81	0.48 (0.45, 0.50)§	<0.001	0.53 (0.50, 0.56)§	<0.001
Yes	186	58 253.04	3.19	99	64 156.19	1.54	0.48 (0.38, 0.62)§	<0.001	0.43 (0.32, 0.59)§	<0.001
Hypertension										
No	2087	560 151.09	3.73	1039	581 923.25	1.79	0.48 (0.44, 0.51)§	<0.001	0.54 (0.49, 0.59)§	<0.001
Yes	4041	1 068 039.58	3.78	2120	1 177 207.11	1.80	0.47 (0.45, 0.50)§	<0.001	0.52 (0.49, 0.56)§	<0.001
Dyslipidemia										
No	2408	617 076.68	3.9	1149	652 186.85	1.76	0.45 (0.42, 0.48)§	<0.001	0.52 (0.47, 0.56)§	<0.001
Yes	3720	1 011 113.99	3.68	2010	1 106 943.51	1.82	0.49 (0.46, 0.52)§	<0.001	0.54 (0.50, 0.57)§	<0.001
CKD										
No	5890	1 556 670.38	3.78	3009	1 672 600.38	1.80	0.47 (0.45, 0.50)§	<0.001	0.53 (0.51, 0.56)§	<0.001
Yes	238	71 520.30	3.33	150	86 529.98	1.73	0.51 (0.41, 0.62)§	<0.001	0.44 (0.34, 0.57)§	<0.001
COPD										
No	5575	1 479 660.70	3.77	2881	1 599 112.69	1.80	0.48 (0.46, 0.50)§	<0.001	0.53 (0.50, 0.56)§	<0.001
Yes	553	148 529.98	3.72	278	160 017.67	1.74	0.46 (0.40, 0.54)§	<0.001	0.52 (0.43, 0.62)§	<0.001
Depression										
No	5769	1 535 066.40	3.76	2956	1 660 950.91	1.78	0.47 (0.45, 0.49)§	<0.001	0.53 (0.50, 0.56)§	<0.001
Yes	359	93 124.27	3.86	203	98 179.45	2.07	0.53 (0.45, 0.63)§	<0.001	0.53 (0.42, 0.65)§	<0.001
Schizophrenia										
No	6071	1 610 741.76	3.77	3121	1 741 836.16	1.79	0.47 (0.45, 0.49)§	<0.001	0.53 (0.50, 0.55)§	<0.001
Yes	57	17 448.91	3.27	38	17 294.20	2.20	0.66 (0.44, 1.00)†	0.048	0.83 (0.50, 1.40)	0.49
Rheumatoid diseases										
No	6019	1 602 198.30	3.76	3117	1 731 050.54	1.80	0.48 (0.46, 0.50)§	<0.001	0.53 (0.50, 0.56)§	<0.001
Yes	109	25 992.38	4.19	42	28 079.82	1.50	0.36 (0.25, 0.51)§	<0.001	0.40 (0.26, 0.61)§	<0.001
Osteoporosis										
No	5849	1 551 350.18	3.77	3029	1 678 442.37	1.80	0.48 (0.46, 0.50)§	<0.001	0.53 (0.50, 0.56)§	<0.001
Yes	279	76 840.49	3.63	130	80 687.99	1.61	0.44 (0.36, 0.54)§	<0.001	0.47 (0.36, 0.60)§	<0.001
Alcoholism										
No	6055	1 601 125.16	3.78	3118	1 730 875.84	1.80	0.47 (0.45, 0.50)§	<0.001	0.53 (0.50, 0.56)§	<0.001
Yes	73	27 065.51	2.7	41	28 254.52	1.45	0.54 (0.37, 0.79)‡	0.001	0.64 (0.40, 1.02)	0.063

(*Continued*)

Table 4. (Continued.)

Variable	POAG in Non-DPP4i User			POAG in DPP4i User			Crude		Adjusted*	
	n	*PY*	*IR*	*n*	*PY*	*IR*	*HR (95% CI)*	P *Value*	*HR (95% CI)*	P *Value*
Nicotine dependence										
No	6033	1 593 951.21	3.78	3106	1 722 631.05	1.80	0.47 (0.45, 0.50)§	<0.001	0.53 (0.50, 0.56)§	<0.001
Yes	95	34 239.47	2.77	53	36 499.31	1.45	0.52 (0.37, 0.72)§	<0.001	0.54 (0.36, 0.81)‡	0.003
Cancer										
No	5871	1 560 193.10	3.76	3035	1 686 750.18	1.80	0.48 (0.46, 0.50)§	<0.001	0.53 (0.50, 0.56)§	<0.001
Yes	257	67 997.57	3.78	124	72 380.18	1.71	0.45 (0.36, 0.55)§	<0.001	0.51 (0.39, 0.66)§	<0.001
Retinopathy										
No	5343	1 504 462.05	3.55	2740	1 601 776.82	1.71	0.48 (0.46, 0.50)§	<0.001	0.54 (0.51, 0.57)§	<0.001
Yes	785	123 728.63	6.34	419	157 353.54	2.66	0.42 (0.37, 0.48)§	<0.001	0.47 (0.41, 0.54)§	<0.001
Myopia										
No	6058	1 615 745.16	3.75	3104	1 744 791.36	1.78	0.47 (0.45, 0.49)§	<0.001	0.53 (0.50, 0.55)§	<0.001
Yes	70	12 445.52	5.62	55	14 339.00	3.84	0.65 (0.46, 0.93)†	0.019	0.74 (0.49, 1.12)	0.154
Cataract										
No	4412	1 294 052.01	3.41	2301	1 390 731.87	1.65	0.48 (0.46, 0.51)§	<0.001	0.54 (0.51, 0.57)§	<0.001
Yes	1716	334 138.67	5.14	858	368 398.49	2.33	0.45 (0.42, 0.49)§	<0.001	0.50 (0.45, 0.55)§	<0.001
Uveitis										
No	6081	1 619 666.37	3.75	3138	1 750 131.58	1.79	0.48 (0.46, 0.50)§	<0.001	0.53 (0.50, 0.56)§	<0.001
Yes	47	8524.30	5.51	21	8998.78	2.33	0.43 (0.26, 0.72)‡	0.001	0.70 (0.38, 1.29)	0.253
DR										
No	5728	1 572 520.62	3.64	2934	1 674 949.10	1.75	0.48 (0.46, 0.50)§	<0.001	0.54 (0.51, 0.57)§	<0.001
Yes	400	55 670.05	7.19	225	84 181.26	2.67	0.38 (0.32, 0.44)§	<0.001	0.43 (0.35, 0.52)§	<0.001
DKD										
No	5716	1 528 295.61	3.74	2899	1 628 602.55	1.78	0.47 (0.45, 0.50)§	<0.001	0.53 (0.50, 0.56)§	<0.001
Yes	412	99 895.07	4.12	260	130 527.81	1.99	0.48 (0.41, 0.56)§	<0.001	0.49 (0.41, 0.59)§	<0.001
DN										
No	5713	1 553 963.27	3.68	2939	1 656 300.80	1.77	0.48 (0.46, 0.50)§	<0.001	0.54 (0.51, 0.57)§	<0.001
Yes	415	74 227.40	5.59	220	102 829.56	2.14	0.38 (0.32, 0.45)§	<0.001	0.43 (0.35, 0.52)§	<0.001
CCI score										
0	5061	1 316 456.12	3.84	2576	1 415 232.20	1.82	0.47 (0.45, 0.50)§	<0.001	0.53 (0.50, 0.56)§	<0.001
1	542	155 780.52	3.48	297	171 588.48	1.73	0.49 (0.43, 0.57)§	<0.001	0.55 (0.47, 0.65)§	<0.001
≥2	525	155 954.04	3.37	286	172 309.67	1.66	0.49 (0.42, 0.56)§	<0.001	0.54 (0.45, 0.64)§	<0.001

CCI = Charlson Comorbidity Index; CI = confidence interval; CKD = chronic kidney disease; COPD = chronic obstructive pulmonary disease; DKD = diabetic kidney disease; DN = diabetic neuropathy; DPP4i = dipeptidyl peptidase-4 inhibitor; DR = diabetic retinopathy; HR = hazard ratio; IR = incidence rate per 1000 person-years; NTD = New Taiwan Dollar; POAG = primary open angle glaucoma; PY = person-years; T2DM = type 2 diabetes mellitus.

*Adjusted HR: adjusted by sex, age, region of dwelling, degree of urbanization, insurance premium, comorbidities, CCI score, and medication.

†*P* value <0.05.

‡*P* < 0.01.

§*P* < 0.001.

cancer, 0.47 (95% CI, 0.41–0.54; $P < 0.001$) for retinopathy, 0.50 (95% CI, 0.45–0.55; $P < 0.001$) for cataract, 0.43 (95% CI, 0.35–0.52; $P < 0.001$) for diabetic retinopathy, 0.49 (95% CI, 0.41–0.59; $P < 0.001$) for diabetic kidney disease, and 0.43 (95% CI, 0.35–0.52; $P < 0.001$) for DN. This association persisted even when compared with users of first-generation diabetes medication (including metformin, sulfonylureas, thiazolidinediones, α-glucosidase inhibitors, and glinide), with aHR ranging from 0.42 to 0.55 across various comparisons (all $P < 0.001$) (Table 5).

DPP4i Use and Risk of NTG

Similarly, DPP4i users demonstrated a lower risk of NTG, with an incidence rate of 0.41 per 1000 person-years compared with 0.74 in non-DPP4i users. The aHR for NTG among DPP4i users was 0.55 (95% CI, 0.50–0.62; $P < 0.001$) (Table 6). Figure 3 illustrates the cumulative incidence curve of NTG, with a consistently lower trend for DPP4i users throughout the study period (log-rank $P < 0.001$).

Discussion

In this large nationwide retrospective cohort study, we found that DPP4i use was associated with a significantly lower risk of developing POAG and NTG in patients with T2DM. This association remained consistent across various subgroups and was evident in cumulative incidence analyses. These findings suggest that DPP4i may confer benefits beyond glycemic control, possibly influencing optic nerve health through multiple mechanisms.

Table 5. Hazard Ratios of POAG among T2DM with Different DM Drug User

Variable	POAG			Crude		Adjusted*	
	n	*PY*	*IR*	*HR (95% CI)*	P *Value*	*HR (95% CI)*	P *Value*
Metformin user	5788	1 564 830.05	3.70	1.00 (reference)	-	1.00 (reference)	-
DPP4i user	3159	1 759 130.36	1.80	0.48 (0.46, 0.50)§	<0.001	0.55 (0.53, 0.58)§	<0.001
SU user	5048	1 249 983.76	4.04	1.00 (reference)	-	1.00 (reference)	-
DPP4i user	3159	1 759 130.36	1.80	0.44 (0.42, 0.46)§	<0.001	0.45 (0.43, 0.47)§	<0.001
TZD user	1518	333 211.13	4.56	1.00 (reference)	-	1.00 (reference)	-
DPP4i user	3159	1 759 130.36	1.80	0.39 (0.37, 0.42)§	<0.001	0.42 (0.39, 0.45)§	<0.001
AGI user	1321	284 857.21	4.64	1.00 (reference)	-	1.00 (reference)	-
DPP4i user	3159	1 759 130.36	1.80	0.39 (0.36, 0.41)§	<0.001	0.43 (0.40, 0.46)§	<0.001
Glinide user	187	55 475.45	3.37	1.00 (reference)	-	1.00 (reference)	-
DPP4i user	3159	1 759 130.36	1.80	0.50 (0.43, 0.58)§	<0.001	0.53 (0.46, 0.62)§	<0.001

AGI = α-glucosidase inhibitor; CI = confidence interval; DM = diabetes mellitus; DPP4i = dipeptidyl peptidase-4 inhibitor; HR = hazard ratio; IR = incidence rate per 1000 person-years; POAG = primary open angle glaucoma; PY = person-years; SU = sulfonylurea; T2DM = type 2 diabetes mellitus; TZD = thiazolidinedione.
*Adjusted HR: adjusted by sex, age, region of dwelling, degree of urbanization, insurance premium, comorbidities, CCI score, and medication.
§$P < 0.001$.

Incretins Action and Beyond

The beneficial effects associated with DPP4i on neurodegenerative diseases are likely multifactorial, encompassing both incretin-dependent and incretin-independent pathways. Previous studies have primarily focused on incretin-related mechanisms. For example, in a rat model of cisplatin-induced neurotoxicity, DPP4i (vildagliptin) significantly increased levels of adenosine monophosphate-activated protein kinase, protein kinase B, and cyclic adenosine monophosphate response element-binding protein—key components of the incretin signaling pathway—thereby alleviating neuroinflammation and cognitive decline.[15] Similarly, Li et al[11] reported that DPP4i (sitagliptin) elevated CNS incretin levels in a rodent model of Parkinson disease, reducing inflammatory markers in the brain.

Li et al further demonstrated that DPP4i, in combination with GLP-1 and glucose-dependent insulinotropic polypeptide, provided superior neuroprotective effects in both rat and human neuronal cultures compared with glucose-dependent insulinotropic polypeptide or GLP-1 alone, suggesting potential synergistic mechanisms beyond incretin signaling.[11] These findings underscore the complex interplay of pathways in modulating inflammation as well as promoting neuroregeneration, which may contribute to the observed association between DPP4i use and glaucoma.[17–27]

Antioxidant Effects and Ocular Connective Tissue Remodeling

Hyperglycemia in diabetic retinas induces excessive mitochondrial production of ROS, leading to the formation of methylglyoxal (MG) and advanced glycation end products (AGEs).[18] In turn, MG and AGEs contribute to mitochondrial dysfunction and oxidative stress, both of which are implicated in the development and progression of glaucoma.[19] Moreover, MG and AGEs increase the stiffness of collagen-rich tissues through protein cross-linking, affecting structures such as the lamina cribrosa and peripapillary sclera. This reduced elasticity compromises the tissue's load-bearing capacity, impairs neurovascular function, and heightens the vulnerability of retinal

Table 6. Hazard Ratios of NTG among T2DM with DPP4i and Non-DPP4i User

Variable	NTG			Crude		Adjusted*	
	n	*PY*	*IR*	*HR (95% CI)*	P *Value*	*HR (95% CI)*	P *Value*
Non-DPP4i user	1220	1 655 251.00	0.74	1.00 (reference)	-	1.00 (reference)	-
DPP4i user	734	1 768 746.13	0.41	0.56 (0.51, 0.61)§	<0.001	0.55 (0.50, 0.62)§	<0.001

CI = confidence interval; DPP4i = dipeptidyl peptidase-4 inhibitor; HR = hazard ratio; IR = incidence rate per 1000 person-years; NTG = normal tension glaucoma; PY = person-years; T2DM = type 2 diabetes mellitus.
*Adjusted HR: adjusted by sex, age, region of dwelling, degree of urbanization, insurance premium, comorbidities, CCI score, and medication.
§$P < 0.001$.

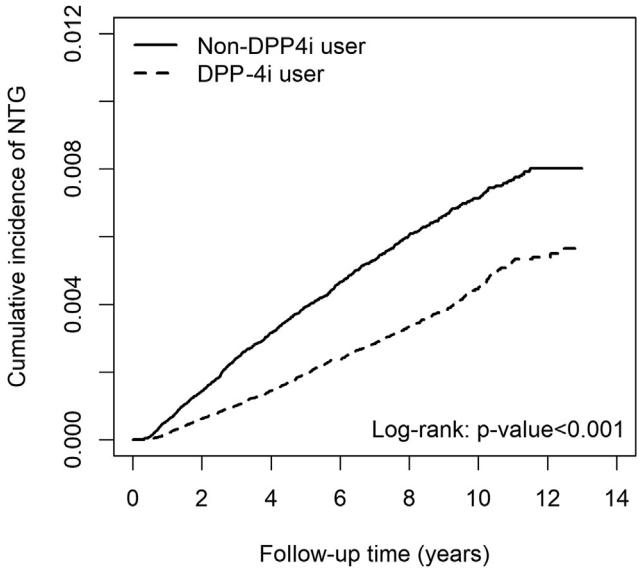


Figure 3. The cumulative incidence curves of NTG in DPP4i and non-DPP4i users. Kaplan–Meier curves showing a consistently lower cumulative incidence of NTG in DPP4i users compared with non-DPP4i users. DPP4i = dipeptidyl peptidase-4 inhibitor; NTG = normal tension glaucoma.

ganglion cells.[20,21] Notably, Hopkins et al[21] argue that tissue stiffness, rather than absolute intraocular pressure or translaminar pressure, drives glaucomatous cupping.

Dipeptidyl peptidase-4 inhibitors exhibit both direct and indirect antioxidant effects by reducing ROS production and disrupting the feedback loop through inhibition of AGEs and their receptor interactions.[22] In diabetic animal models, DPP4is (linagliptin) significantly decreased ROS and MG levels. The treatment also preserved cell numbers in the ganglion cell layer, maintaining levels comparable to nondiabetic controls.[12] These antioxidative properties, combined with their ability to remodel ocular connective tissue, may explain the associations observed in our study. This suggests that DPP4i may mitigate some of the oxidative toxicity as well as structural changes associated with glaucoma progression.

Mitigating Glutamate-Induced Excitotoxicity

Excessive glutamate, the primary excitatory neurotransmitter in the CNS, can cause excitotoxicity, contributing to the pathogenesis of various neurodegenerative conditions, including Alzheimer disease, Parkinson disease, and glaucoma.[23–25] Glutamate is typically cleared by astrocytic transporters such as glutamate-aspartate transporter (GLAST), which converts it to nontoxic glutamine via glutamine synthetase.[23,25] However, diabetic retinas show reduced GLAST expression and elevated glutamate levels, exacerbating excitotoxic damage.[26,27]

Dipeptidyl peptidase-4 inhibitors have demonstrated efficacy in reducing glutamate concentrations and preserving GLAST levels in diabetic models. For instance, topical DPP4i (saxagliptin and sitagliptin) treatment significantly reduced retinal glutamate levels to those comparable with controls, and prevented GLAST downregulation in diabetic rat model, highlighting its potential neuroprotective effects independent of glycemic control.[26] These mechanisms may contribute to the reduced risk of glaucoma observed among DPP4i users in our study.

The strengths of this study include its large sample size, robust matching methods, and comprehensive consideration of potential confounders. However, several limitations should be acknowledged. Our study relied on administrative claims data, which lacked detailed clinical metrics such as intraocular pressure, family history, genetic predisposition, and corneal thickness. Additionally, researchers using health insurance databases often encounter coding bias. This could potentially influence the identification of cases. Future prospective studies are warranted to explore the mechanisms underlying DPP4i's effects on glaucoma risk.

In conclusion, our nationwide cohort study provides evidence that DPP4i use is associated with a lower risk of developing POAG and NTG in adults with T2DM. These findings highlight the potential benefits of DPP4i for glaucoma prevention strategies in clinical practice.

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Abbreviations and Acronyms:
AGE = advanced glycation end product; **aHR** = adjusted hazard ratio; **CI** = confidence interval; **CM** = Clinical Modification; **DN** = diabetic neuropathy; **DPP4i** = dipeptidyl peptidase-4 inhibitor; **GLAST** = glutamate-aspartate transporter; **GLP-1 RA** = glucagon-like peptide-1 receptor agonist; **ICD** = International Classification of Diseases; **MG** = methylglyoxal; **NTG** = normal tension glaucoma; **POAG** = primary open-angle glaucoma; **ROS** = reactive oxygen species; **T2DM** = type 2 diabetes mellitus.

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